Lab 1 - Best to date F24

$$\underbrace{\dot{x} + x = 1,}\quad \underbrace{x(0) = 0}$$

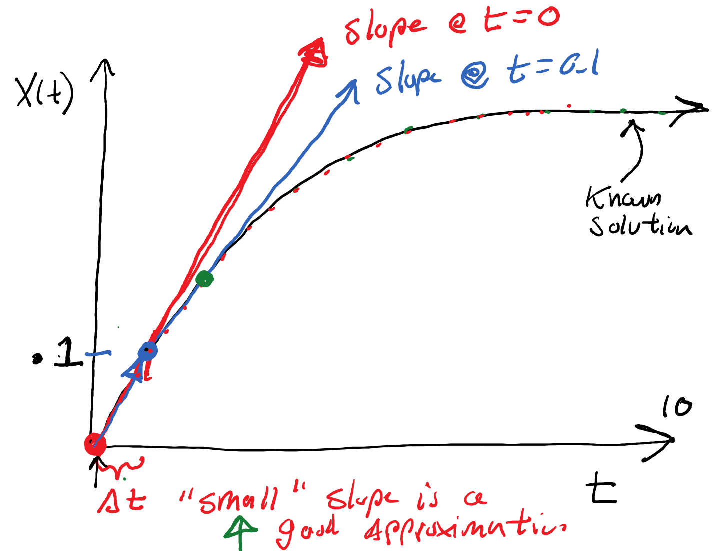
$$\dot{x}(t) + x(t) = 1$$

$$\boxed{\dot{x}(t) = 1 - x(t)}$$

@ $t = 0$

$$\dot{x}(0) = 1 - x(0) = 1 - 0 = 1 = \frac{dx(0)}{dt}$$

↳ slope of $x(t)$ @ $t=0$



$\Delta t = 0.1$

Use slope to calculate next point

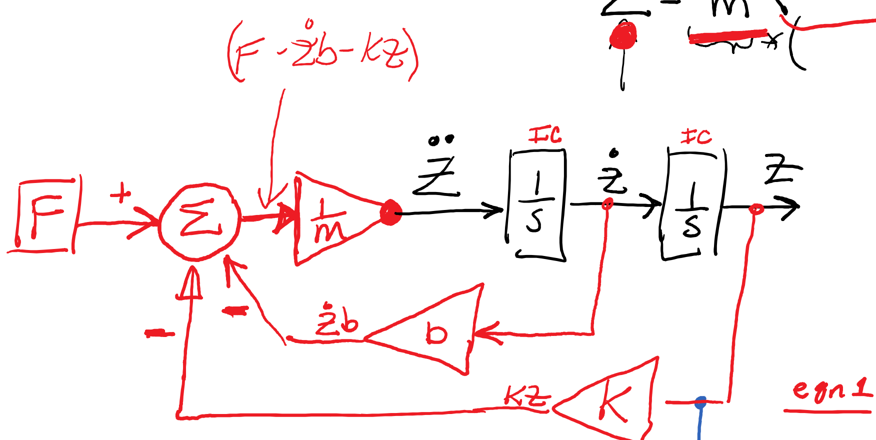
$$x(0) + \text{slope}(0) * \Delta t = 0 + 1 * (0.1) = 0.1$$

↓ repeat

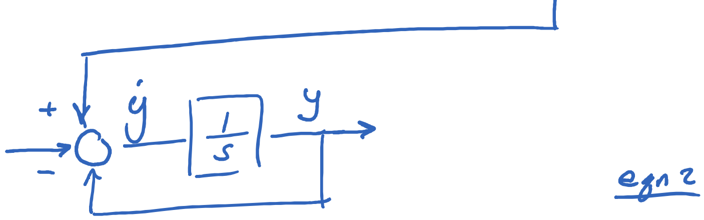
$$m\ddot{z} + b\dot{z} + kz = F$$

Solve highest derivative

$$\ddot{z} = \frac{1}{m}\left(F - b\dot{z} - kz\right)$$



$$\ddot{z} = \frac{1}{m} * \left(F - b\dot{z} - kz\right)$$



(1) $m\ddot{z} + b\dot{z} + kz = F$

(2) $\dot{y} + y = z \Rightarrow \dot{y} = z - y$